

# 04 Cross-Validation Selection Error Rate

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## Goal (cross-validation)

For each sample size on 50, 100, ..., 1000 (step 50), we run **10-fold** cross-validation for a **linear** model (M1) and **degree-3 polynomial** (M3), select the model with lower mean CV MSE (tie  $\rightarrow$  linear), and report the Monte Carlo **selection error rate** vs **n**, with **500** replications per **n**.

## Linear truth

### Simulation settings

Table 1: Simulation settings (cross-validation, linear truth).

| setting                        | value                            |
|--------------------------------|----------------------------------|
| seed                           | 2026                             |
| Monte Carlo replications per n | 500                              |
| Sample sizes (grid)            | 50, 100, ..., 1000 (step 50)     |
| truth type                     | linear                           |
| beta0                          | 1                                |
| beta1                          | 2                                |
| beta2                          | NA                               |
| beta3                          | NA                               |
| noise type                     | gaussian                         |
| sigma                          | 2                                |
| CV folds                       | 10                               |
| selection rule                 | minimum average validation error |

## Error rate summary

Table 2: Monte Carlo mean selection-error rate (truth is linear; error = poly chosen) and approximate MC standard error.

| n    | error_rate | mc_se   |
|------|------------|---------|
| 50   | 0.138      | 0.01542 |
| 100  | 0.156      | 0.01623 |
| 150  | 0.184      | 0.01733 |
| 200  | 0.142      | 0.01561 |
| 250  | 0.158      | 0.01631 |
| 300  | 0.152      | 0.01606 |
| 350  | 0.126      | 0.01484 |
| 400  | 0.160      | 0.01640 |
| 450  | 0.156      | 0.01623 |
| 500  | 0.150      | 0.01597 |
| 550  | 0.170      | 0.01680 |
| 600  | 0.150      | 0.01597 |
| 650  | 0.138      | 0.01542 |
| 700  | 0.176      | 0.01703 |
| 750  | 0.152      | 0.01606 |
| 800  | 0.180      | 0.01718 |
| 850  | 0.138      | 0.01542 |
| 900  | 0.200      | 0.01789 |
| 950  | 0.132      | 0.01514 |
| 1000 | 0.180      | 0.01718 |

## Error rate vs sample size

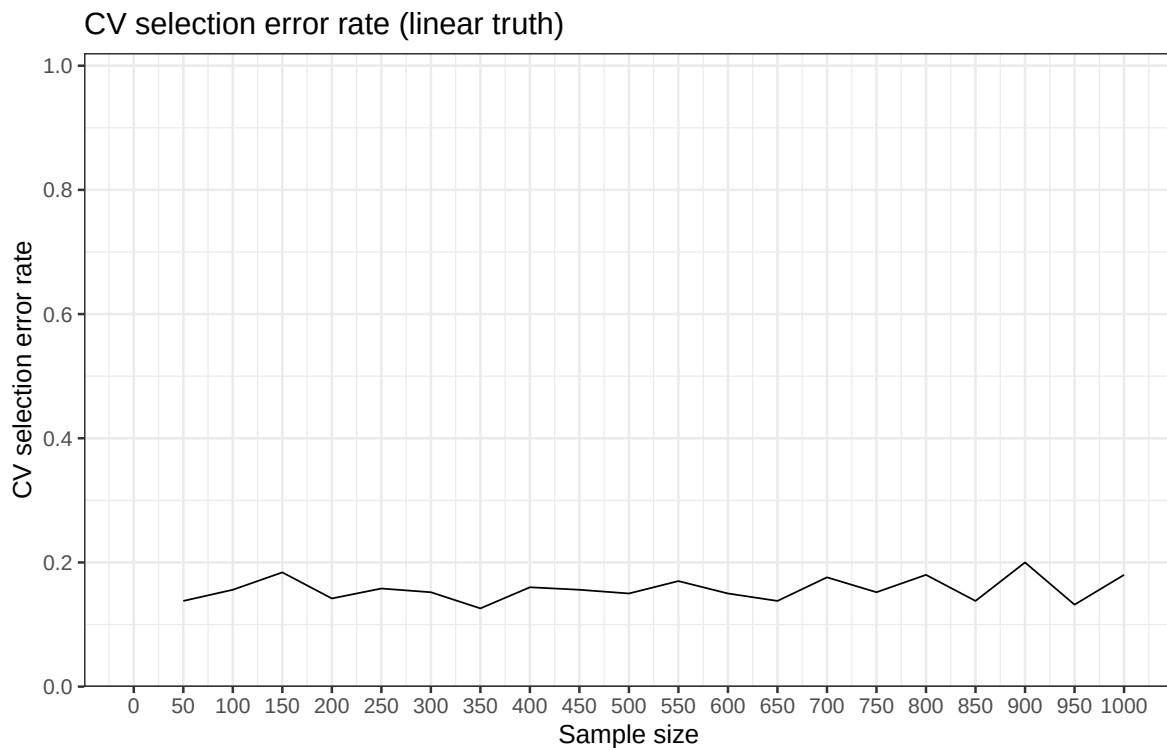


Figure 1: CV selection error rate under linear truth.

## Polynomial truth

### Simulation settings

Table 3: Simulation settings (cross-validation, polynomial truth).

| setting                        | value                        |
|--------------------------------|------------------------------|
| seed                           | 2026                         |
| Monte Carlo replications per n | 500                          |
| Sample sizes (grid)            | 50, 100, ..., 1000 (step 50) |
| truth type                     | polynomial                   |
| beta0                          | 1                            |
| beta1                          | 2                            |
| beta2                          | 1                            |

| setting        | value                            |
|----------------|----------------------------------|
| beta3          | 1                                |
| noise type     | gaussian                         |
| sigma          | 2                                |
| CV folds       | 10                               |
| selection rule | minimum average validation error |

## Error rate summary

Table 4: Monte Carlo mean selection-error rate (truth is polynomial; error = linear chosen) and approximate MC standard error.

| n    | error_rate | mc_se   |
|------|------------|---------|
| 50   | 0.028      | 0.00738 |
| 100  | 0.000      | 0.00000 |
| 150  | 0.000      | 0.00000 |
| 200  | 0.000      | 0.00000 |
| 250  | 0.000      | 0.00000 |
| 300  | 0.000      | 0.00000 |
| 350  | 0.000      | 0.00000 |
| 400  | 0.000      | 0.00000 |
| 450  | 0.000      | 0.00000 |
| 500  | 0.000      | 0.00000 |
| 550  | 0.000      | 0.00000 |
| 600  | 0.000      | 0.00000 |
| 650  | 0.000      | 0.00000 |
| 700  | 0.000      | 0.00000 |
| 750  | 0.000      | 0.00000 |
| 800  | 0.000      | 0.00000 |
| 850  | 0.000      | 0.00000 |
| 900  | 0.000      | 0.00000 |
| 950  | 0.000      | 0.00000 |
| 1000 | 0.000      | 0.00000 |

## Error rate vs sample size

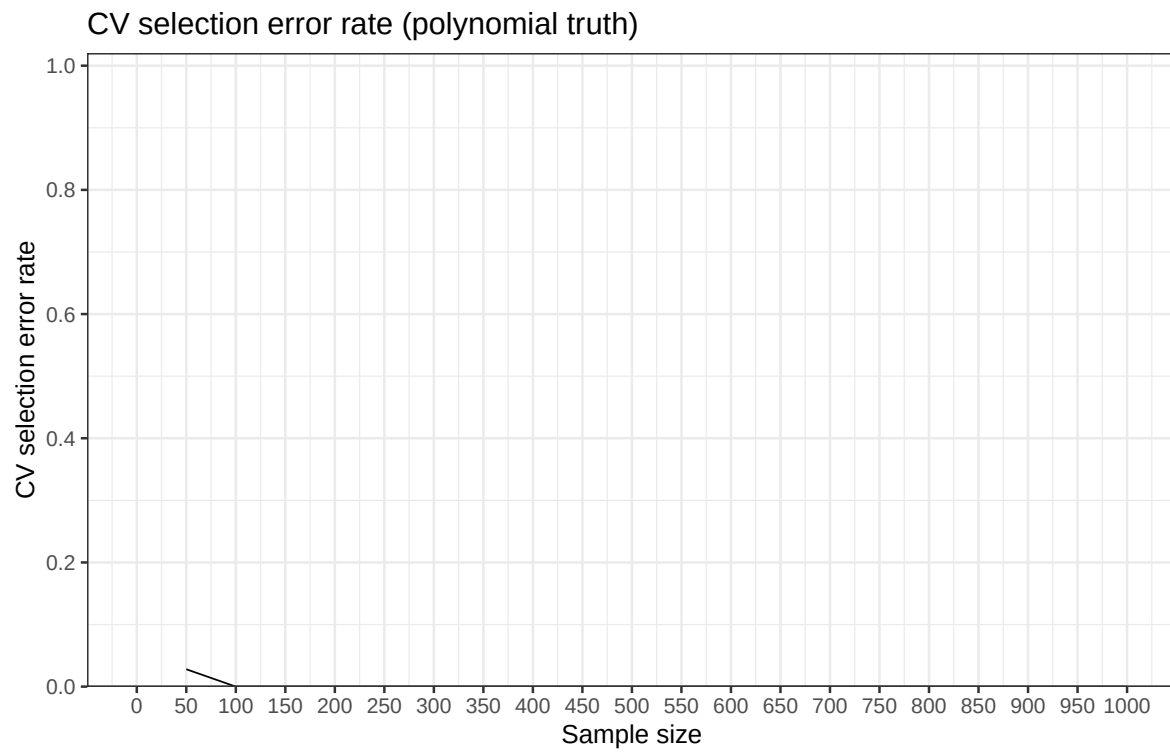


Figure 2: CV selection error rate under polynomial truth.